

Stéphane Karasiewicz

Climate Pipeline Partner · Senior Data Engineer for Parametric Insurance

Outreau, France · Remote worldwide ·   native

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I help parametric insurers ship climate scoring pipelines in 8 to 16 weeks — bridging climate science and DevOps, without recruiting in-house.

Experience

Skaraz Data — Climate Pipeline Partner & Founder

Sep 2023 → Present · Remote

Senior data engineering for parametric insurance climate pipelines.

- › Audit, build, and operate production climate scoring pipelines (drought, heat stress, precipitation, water balance)
- › 3-tier engagement: Discovery (€5-7k, 2 weeks) · Partner (€25-50k, 8-16 weeks) · Retainer (€2-4k/month)
- › Code Quality Guarantee: any major bug in production on shipped code fixed for free within 30 days of handover
- › Available: 2-3 new engagements per quarter

Raincoat LLC — Senior Data Scientist (Freelance contract)

Sep 2023 → Jan 2026 · Remote

Parametric insurtech (Seed-stage, San Juan, Puerto Rico) · Climate risk technology

- › Built and operated end-to-end ML pipelines for **4 climate perils** (drought, heat stress, precipitation, water balance) across **10+ countries** in LatAm, SEA, and Africa
- › Engineered automated data management for **2 TB+ satellite & reanalysis data** (CHIRPS, IMERG, ERA5-Land, Copernicus, MERRA-2)
- › Created **2 production pipelines from scratch** (heat stress + water balance); delivered a **modular architecture proposal** as scoped technical roadmap for next-perils scaling, transferred to the team at contract end
- › Ran dev/ops solo for production pipelines at end of contract; built 5 country-level operational dashboards
- › Tools: Python, xarray, GDAL, Docker, PostgreSQL, scikit-learn

IFREMER — Marine Remote Sensing Researcher

Feb 2018 → Jul 2023 · Boulogne-sur-Mer & Brest, France

French National Institute for Ocean Science · 5+ years in satellite remote sensing for marine ecology

- › Operated production R pipelines processing 500 GB+ satellite & in-situ data; reduced manual effort by 50 hours/month
- › Authored 2 CRAN packages (BDAIgo, subniche) with **8,000+ cumulative downloads** and 27+ citations
- › 5 peer-reviewed publications on environmental modeling using MODIS, Sentinel-2/3, and reanalysis data
- › Mentored 6 researchers and Master's students on R, Git, and reproducible research

Technical stack

Climate data — CHIRPS · IMERG · ERA5-Land · Copernicus · MERRA-2 · MODIS · Sentinel

Engineering — Python (expert) · xarray · GDAL · Docker · PostgreSQL · Airflow · CI/CD

Languages — Python · R (2 CRAN packages) · SQL · Bash

Methods — Production data pipelines · Geospatial analysis · Time-series · Multivariate stats

Education

PhD Marine Ecology — Université de Lille, France (2014-2017) · Phytoplankton evolution — Novel WitOMI statistical framework for ecological niches

M.Sc. Marine Ecology, Stockholm University (2012-2014) · **B.Sc. Ecosystem Science**, Open University, England (2010-2012)

Publications & Open Source

CRAN R packages

- › **subniche** — Ecological niche modeling using OMI framework · 5,000+ downloads · 15+ citations
- › **BDAIgo** — Bloom detection in time-series, extracts 22 phenological variables · 3,000+ downloads · 12+ citations

Featured publication

- › Karasiewicz S., Chapelle A., Bacher C., Soudant D. (2020). *Harmful algae niche responses*. **Harmful Algae**, 98:101785.
<https://doi.org/10.1016/j.hal.2020.101785>

5 additional peer-reviewed publications at [ResearchGate](https://www.researchgate.net/publications/354888888).

Languages

French — Native · English — Native (dual citizenship)

Available — 2-3 new engagements per quarter

Best contact: LinkedIn DM with “reality” for the free Climate Pipeline Reality Check PDF.